

Rakesh gurung

23048994

Task 1

```
1  using System;
2
3  public class BankAccount
4  {
5      // Private fields
6      private string accountNumber;
7      private double balance;
8
9      // Public property: AccountNumber (GET only)
10     public string AccountNumber
11     {
12         get { return accountNumber; }
13     }
14
15     // Public property: Balance (GET only, private SET)
16     public double Balance
17     {
18         get { return balance; }
19         private set
20         {
21             if (value <= 0)
22             {
23                 throw new ArgumentException("Balance cannot be zero or negative");
24             }
25             balance = value;
26         }
27     }
28
29     // Constructor
30     public BankAccount(string accNo, double initialBalance)
31     {
32         accountNumber = accNo;
33
34         if (initialBalance <= 0)
35         {
36             throw new ArgumentException("Initial balance must be greater than 0.");
37         }
38
39         balance = initialBalance;
40     }
41
42     // Deposit method
43     public void Deposit(double amount)
44     {
45         if (amount <= 0)
46         {
47             Console.WriteLine("Deposit amount must be greater than zero.");
48             return;
49         }
50
51         balance += amount;
52         Console.WriteLine($"Successfully deposited: Rs {amount}");
53     }
54
55     // Withdraw method
56     public void Withdraw(double amount)
57     {
58         if (amount <= 0)
59         {
60             Console.WriteLine("Withdrawal amount must be greater than zero.");
61             return;
62         }
63
64         if (amount > balance)
65         {
66             Console.WriteLine("Insufficient balance.");
67             return;
68         }
69
70         balance -= amount;
71         Console.WriteLine($"Successfully withdrawn: Rs {amount}");
72     }
73 }
74
```

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```
1  using System;
2
3  class Program
4  {
5      static void Main(string[] args)
6      {
7          // Create object
8          BankAccount bankAccount = new BankAccount("1234", 5000);
9
10         // Display account number & balance using properties
11         Console.WriteLine("Account Number: " + bankAccount.AccountNumber);
12         Console.WriteLine("Initial Balance: Rs " + bankAccount.Balance);
13
14         // Deposit and withdraw
15         bankAccount.Deposit(2000);
16         bankAccount.Withdraw(1000);
17
18         // Remaining balance
19         Console.WriteLine("Remaining Balance: Rs " + bankAccount.Balance);
20     }
21 }
22
```

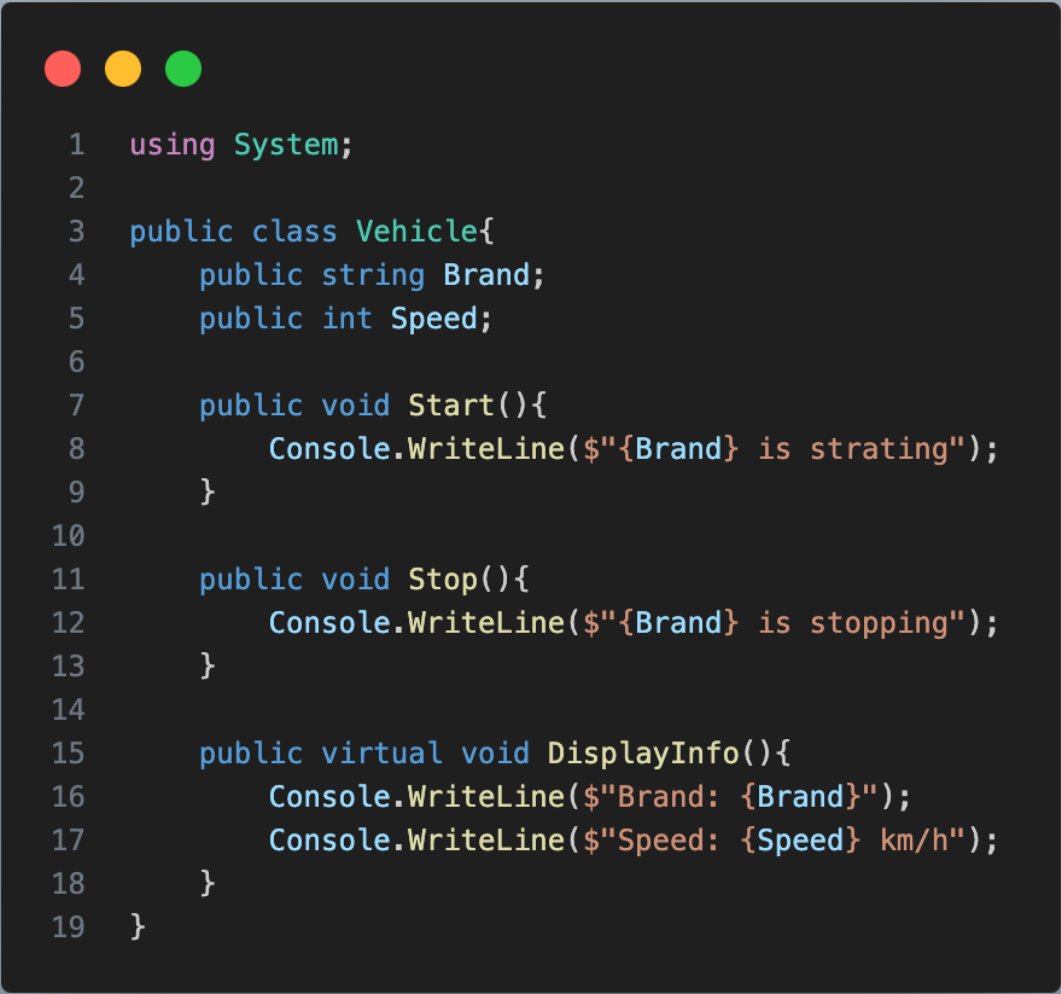
PROBLEMS OUTPUT TERMINAL ...  zsh   ... ^ X

```
● rakeshgurung@Rakeshs-MacBook-Air workshop5 % dotnet run
Account Number: 1234
Initial Balance: Rs 5000
Successfully deposited: Rs 2000
Successfully withdrawn: Rs 1000
Remaining Balance: Rs 6000
```

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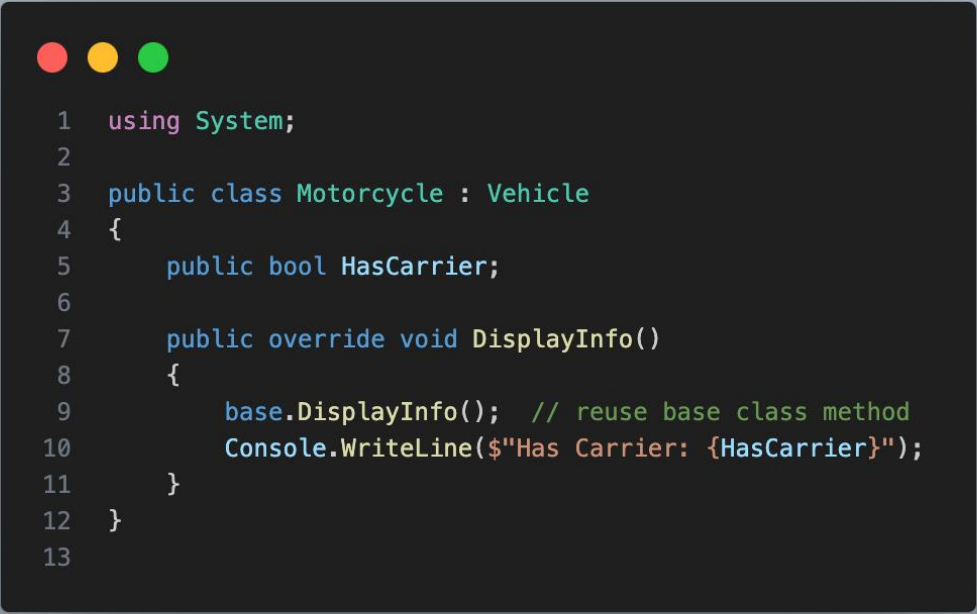
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Task 2

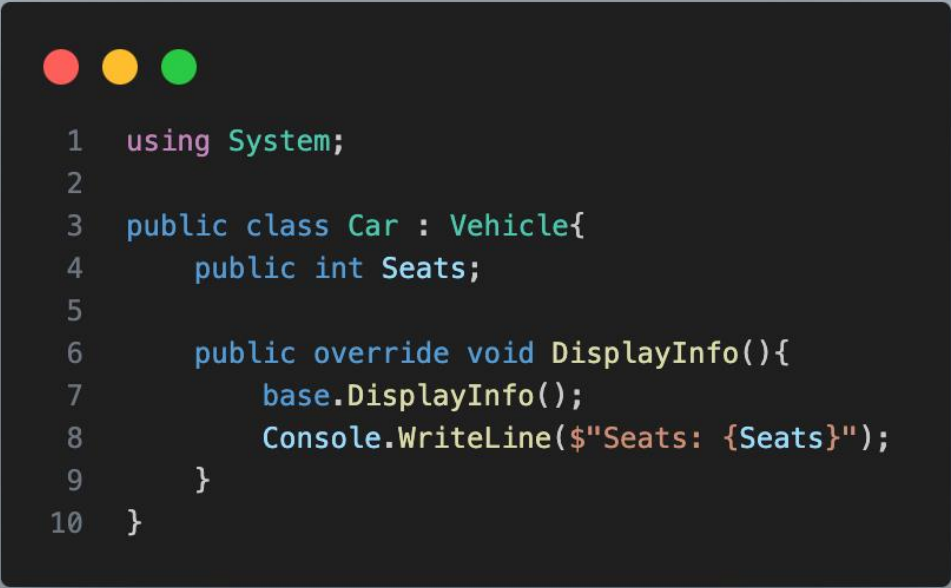


```
1  using System;
2
3  public class Vehicle{
4      public string Brand;
5      public int Speed;
6
7      public void Start(){
8          Console.WriteLine($"{Brand} is strating");
9      }
10
11     public void Stop(){
12         Console.WriteLine($"{Brand} is stopping");
13     }
14
15     public virtual void DisplayInfo(){
16         Console.WriteLine($"Brand: {Brand}");
17         Console.WriteLine($"Speed: {Speed} km/h");
18     }
19 }
```

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```
1  using System;
2
3  public class Motorcycle : Vehicle
4  {
5      public bool HasCarrier;
6
7      public override void DisplayInfo()
8      {
9          base.DisplayInfo(); // reuse base class method
10         Console.WriteLine($"Has Carrier: {HasCarrier}");
11     }
12 }
13
```



```
1  using System;
2
3  public class Car : Vehicle{
4      public int Seats;
5
6      public override void DisplayInfo(){
7          base.DisplayInfo();
8          Console.WriteLine($"Seats: {Seats}");
9      }
10 }
```

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```
1
2  using System;
3
4  class Program
5  {
6      static void Main(string[] args)
7      {
8          // Object of Car
9          Car car = new Car();
10         car.Brand = "Toyota";
11         car.Speed = 180;
12         car.Seats = 5;
13
14         // Object of Motorcycle
15         Motorcycle bike = new Motorcycle();
16         bike.Brand = "Yamaha";
17         bike.Speed = 120;
18         bike.HasCarrier = true;
19
20         // Demonstrate methods and code reuse
21         Console.WriteLine("=== Car Info ===");
22         car.Start();
23         car.DisplayInfo();
24         car.Stop();
25
26         Console.WriteLine("\n=== Motorcycle Info ===");
27         bike.Start();
28         bike.DisplayInfo();
29         bike.Stop();
30     }
31 }
```

PROBLEMS OUTPUT TERMINAL ... zsh + - [] [] ... ^

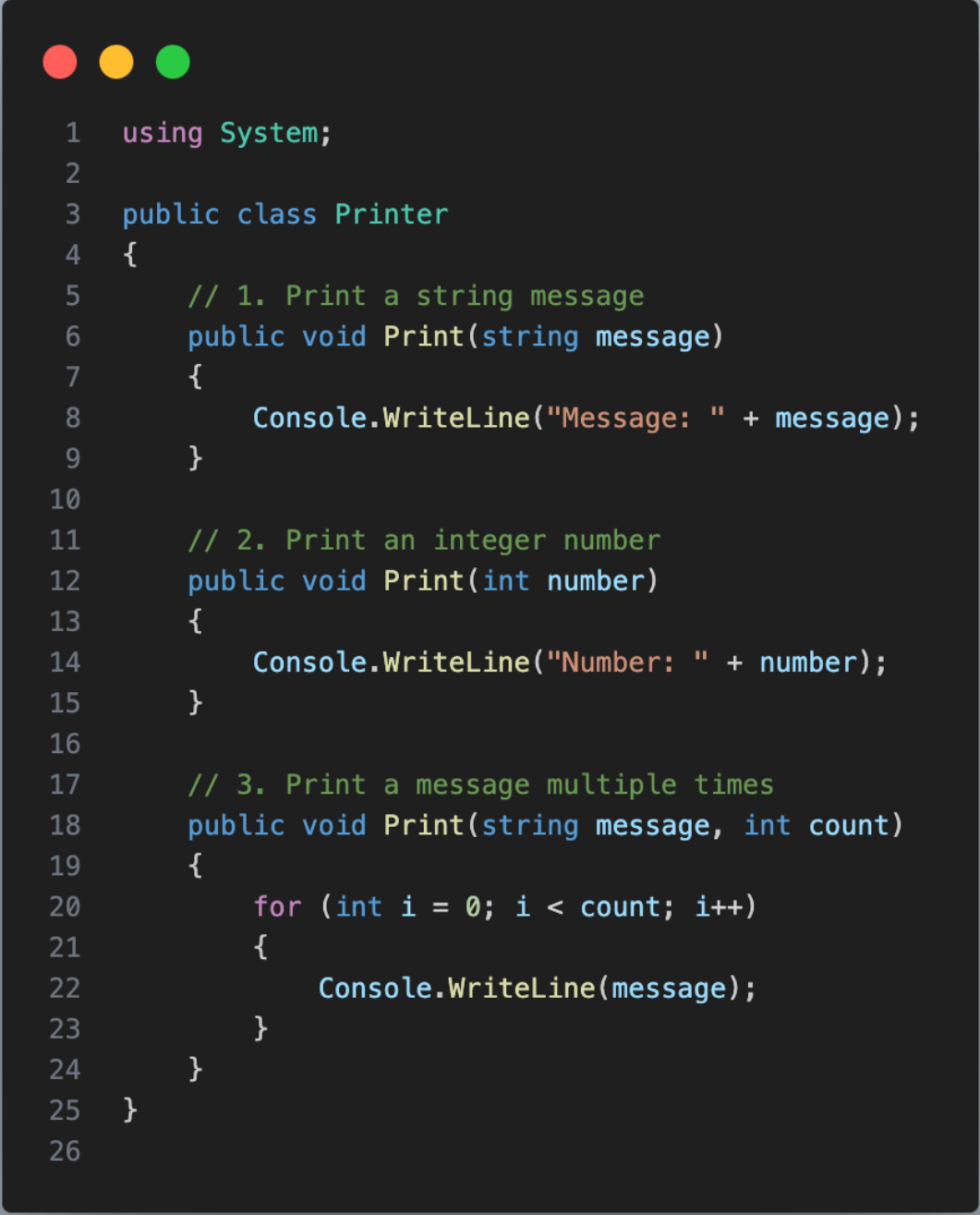
```
=== Car Info ===
Toyota is strating
Brand: Toyota
Speed: 180 km/h
Seats: 5
Toyota is stopping

=== Motorcycle Info ===
Yamaha is strating
Brand: Yamaha
Speed: 120 km/h
Has Carrier: True
Yamaha is stopping
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```

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Task 3



```
1  using System;
2
3  public class Printer
4  {
5      // 1. Print a string message
6      public void Print(string message)
7      {
8          Console.WriteLine("Message: " + message);
9      }
10
11     // 2. Print an integer number
12     public void Print(int number)
13     {
14         Console.WriteLine("Number: " + number);
15     }
16
17     // 3. Print a message multiple times
18     public void Print(string message, int count)
19     {
20         for (int i = 0; i < count; i++)
21         {
22             Console.WriteLine(message);
23         }
24     }
25 }
26
```

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```
1  class Program
2  {
3      static void Main(string[] args)
4      {
5          Printer printer = new Printer();
6
7          // Calling Print(string)
8          printer.Print("Hello, this is a message!");
9
10         // Calling Print(int)
11         printer.Print(100);
12
13         // Calling Print(string, int)
14         printer.Print("I will be printed 3 times!", 3);
15
16     }
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
zsh ⚠ + - □ 🗑 ... ^ ×

● rakeshgurung@Rakeshs-MacBook-Air workshop5 % dotnet run
/Users/rakeshgurung/Documents/3rd Year/AD/workshop5/Vehicle.cs(4,19): warning CS8618: Non-nullable field 'Brand' must contain a non-null value when exiting constructor. Consider adding the 'required' modifier or declaring the field as nullable.
Message: Hello, this is a message!
Number: 100
I will be printed 3 times!
I will be printed 3 times!
I will be printed 3 times!
○ rakeshgurung@Rakeshs-MacBook-Air workshop5 %
```

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Task 4

```
1  using System;
2
3  public class Teacher
4  {
5      public string Name;
6
7      public virtual void Teaching()
8      {
9          Console.WriteLine("Teacher teaches in English.");
10     }
11
12     // This method CANNOT be overridden because it is NOT virtual
13     public void SalaryInfo()
14     {
15         Console.WriteLine("Salary information cannot be changed.");
16     }
17 }
18
```

```
1  using System;
2
3  public class NepaliTeacher : Teacher
4  {
5      public override void Teaching()
6      {
7          Console.WriteLine("Nepali Teacher teaches in Nepali language.");
8      }
9  }
10
```

```
1  using System;
2
3  public class EnglishTeacher : Teacher
4  {
5
6  }
7
```

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```
1  class Program
2  {
3      static void Main(string[] args)
4      {
5          // Nepali teacher object
6          NepaliTeacher nt = new NepaliTeacher();
7          nt.Name = "Ravi Mahrajan";
8          Console.WriteLine("Teacher: " + nt.Name);
9          nt.Teaching();
10         nt.SalaryInfo();
11
12         Console.WriteLine();
13
14         // English teacher object
15         EnglishTeacher et = new EnglishTeacher();
16         et.Name = "Steve Rai";
17         Console.WriteLine("Teacher: " + et.Name);
18         et.Teaching();    // Uses base method
19         et.SalaryInfo(); // Cannot be overridden
20
21         Console.ReadLine();
22     }
23 }
24
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

dotnet ⚠ + ▢ 🗑 ... ^ ×

```
○ rakeshgurung@Rakeshs-MacBook-Air workshop5 % dotnet run
Teacher: Ravi Mahrajan
Nepali Teacher teaches in Nepali language.
Salary information cannot be changed.

Teacher: Steve Rai
Teacher teaches in English.
Salary information cannot be changed.
```

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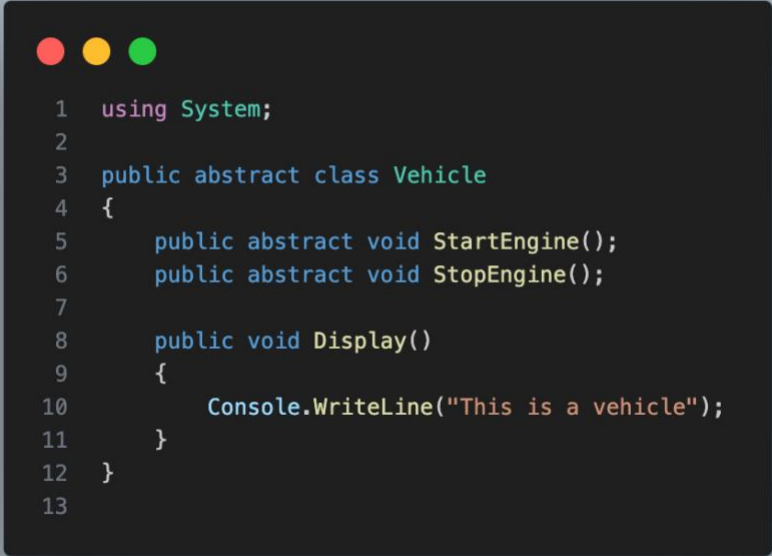
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Task 5

```
1  using System;
2
3  public class Bike : Vehicle
4  {
5      public override void StartEngine()
6      {
7          Console.WriteLine("Bike engine started.");
8      }
9
10     public override void StopEngine()
11     {
12         Console.WriteLine("Bike engine stopped.");
13     }
14 }
15
```

```
1  using System;
2
3  public class Car : Vehicle
4  {
5      public override void StartEngine()
6      {
7          Console.WriteLine("Car engine started.");
8      }
9
10     public override void StopEngine()
11     {
12         Console.WriteLine("Car engine stopped.");
13     }
14 }
15
```

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```
1  using System;
2
3  public abstract class Vehicle
4  {
5      public abstract void StartEngine();
6      public abstract void StopEngine();
7
8      public void Display()
9      {
10         Console.WriteLine("This is a vehicle");
11     }
12 }
13
```

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```
1  class Program
2  {
3      static void Main(string[] args)
4      {
5          Car car = new Car();
6          Bike bike = new Bike();
7
8          Console.WriteLine("CAR:");
9          car.Display();
10         car.StartEngine();
11         car.StopEngine();
12
13         Console.WriteLine();
14
15         Console.WriteLine("BIKE:");
16         bike.Display();
17         bike.StartEngine();
18         bike.StopEngine();
19
20         Console.ReadLine();
21     }
22 }
23
24
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

dotnet + - [] [] ... ^ x

○ rakeshgurung@Rakeshs-MacBook-Air workshop5 % dotnet run

CAR:
This is a vehicle
Car engine started.
Car engine stopped.

BIKE:
This is a vehicle
Bike engine started.
Bike engine stopped.

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Task 6

```
1  using System;
2
3  public abstract class ElectronicDevice
4  {
5      // Private fields
6      private string brand;
7      private double price;
8
9      // Encapsulated public properties
10     public string Brand
11     {
12         get { return brand; }
13         set { brand = value; }
14     }
15
16     public double Price
17     {
18         get { return price; }
19         set
20         {
21             if (value > 0)
22                 price = value;
23             else
24                 price = 0;    // Validation example
25         }
26     }
27
28     // Constructor
29     public ElectronicDevice(string brand, double price)
30     {
31         this.brand = brand;
32         this.price = price;
33     }
34
35     // Abstract method to be implemented by derived classes
36     public abstract void ShowInfo();
37 }
38
```

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```
1  using System;
2
3  public class Laptop : ElectronicDevice
4  {
5      public bool BatteryOn { get; set; }
6
7      public Laptop(string brand, double price)
8          : base(brand, price)
9      {
10         BatteryOn = false;
11     }
12
13     // Child-specific method
14     public void TurnOnBattery()
15     {
16         BatteryOn = true;
17         Console.WriteLine("Laptop battery is now ON.");
18     }
19
20     // Override abstract method
21     public override void ShowInfo()
22     {
23         Console.WriteLine("----- LAPTOP DETAILS -----");
24         Console.WriteLine("Brand: " + Brand);
25         Console.WriteLine("Price: $" + Price);
26         Console.WriteLine("Battery Status: " + (BatteryOn ? "On" : "Off"));
27         Console.WriteLine("-----");
28     }
29 }
30
```

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```
1  using System;
2
3  public class Smartphone : ElectronicDevice
4  {
5      public bool CameraEnabled { get; set; }
6
7      public Smartphone(string brand, double price)
8          : base(brand, price)
9      {
10         CameraEnabled = false;
11     }
12
13     // Child-specific method
14     public void EnableCamera()
15     {
16         CameraEnabled = true;
17         Console.WriteLine("Smartphone camera is now ENABLED.");
18     }
19
20     // Override abstract method
21     public override void ShowInfo()
22     {
23         Console.WriteLine("----- SMARTPHONE DETAILS -----");
24         Console.WriteLine("Brand: " + Brand);
25         Console.WriteLine("Price: $" + Price);
26         Console.WriteLine("Camera Enabled: " + (CameraEnabled ? "Yes" : "No"));
27         Console.WriteLine("-----");
28     }
29 }
30
```

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```
1  using System;
2  using System.Collections.Generic;
3
4  public class ElectronicsStore
5  {
6      private List<ElectronicDevice> devices;
7
8      public ElectronicsStore()
9      {
10         devices = new List<ElectronicDevice>();
11     }
12
13     // Add a new device
14     public void AddDevice(ElectronicDevice device)
15     {
16         devices.Add(device);
17         Console.WriteLine(device.Brand + " added to store.");
18     }
19
20     // Remove a device
21     public void RemoveDevice(ElectronicDevice device)
22     {
23         if (devices.Contains(device))
24         {
25             devices.Remove(device);
26             Console.WriteLine(device.Brand + " removed from store.");
27         }
28         else
29         {
30             Console.WriteLine("Device not found in store.");
31         }
32     }
33
34     // Display details for all devices
35     public void ShowAllDeviceDetails()
36     {
37         Console.WriteLine("\n=== STORE DEVICE DETAILS ===");
38
39         foreach (ElectronicDevice device in devices)
40         {
41             device.ShowInfo(); // Polymorphism call
42
43             // Downcasting to access child-specific behaviors
44             if (device is Laptop laptop)
45             {
46                 laptop.TurnOnBattery();
47             }
48             else if (device is Smartphone phone)
49             {
50                 phone.EnableCamera();
51             }
52
53             Console.WriteLine();
54         }
55     }
56 }
57
```

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```
1  class Program
2  {
3      static void Main(string[] args)
4      {
5          // Create store object
6          ElectronicsStore store = new ElectronicsStore();
7
8          // Create devices
9          Laptop laptop = new Laptop("Dell", 1200.00);
10         Smartphone phone = new Smartphone("Samsung", 999.99);
11
12         // Add devices to store
13         store.AddDevice(laptop);
14         store.AddDevice(phone);
15
16         // Show details and call child methods
17         store.ShowAllDeviceDetails();
18
19         Console.ReadLine();
20     }
21 }
22
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

dotnet ⚠ + ▢ 🗑 ... ^ ×

○ rakeshgurung@Rakeshs-MacBook-Air workshop5 % dotnet run

Dell added to store.
Samsung added to store.

=== STORE DEVICE DETAILS ===

----- LAPTOP DETAILS -----

Brand: Dell
Price: \$1200
Battery Status: Off

Laptop battery is now ON.

----- SMARTPHONE DETAILS -----

Brand: Samsung
Price: \$999.99
Camera Enabled: No

Smartphone camera is now ENABLED.